A Review of Regenerative Braking in Electric Vehicles

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***Abstract*: Growing trends in electrical vehicle Industry initiates us for advance techniques to be developed for enhancing the efficiency of driving. Regenerative braking is about extracting the kinetic energy from the wheels which gets wasted as heat and friction in conventional braking. It is more efficient for vehicles moving at higher speeds. The improvement is done by using flywheel, ultra-capacitor, advanced power electronic converter and efficient energy storage systems. The regenerative braking improves the driving range around 16.25%. Also, the vehicles braking time is reduced. The advance control algorithms like fuzzy logic improves the energy savings in electric vehicle.**

***Keywords:* Regenerative braking system, EV, BLDC motor, brake controller, Energy Recovery Mechanism.**

I. INTRODUCTION

In recent years green Energy has become the need for environmental protection. The electric vehicles produce zero emission to the atmosphere thus helps in reducing air pollution caused by transportation system. The main problem for electrical vehicle is with the driving range. Due to limited space and weight of the batteries, the amount of stored energy carried by EV is less. Thus, regenerative braking helps in recovering the energy from the wheels which gets wasted in traditional braking system. The regenerative braking improves the driving range by 16.25% [3].

When the motor speed reaches above rated speed especially travelling downhill, large amount of back emf is produced in the motor windings. During this regenerative braking will recover maximum energy from the motor. The motor with large winding inductance and low winding resistance favor's regeneration.

In earlier years, electric vehicles are driven by induction motor. Nowadays, Brushless dc motors(BLDC) are used for driving EV's. Though BLDC is having complex control but with the help of power electronics, it has become easier. It has many advantages over induction motor and brushed dc motor like good efficiency and dynamic response, simple structure, greater speed range, robust structure, higher starting torque and noise free operation. It is having nearly constant torque profile. The merits of regeneration are as follows:

1. Better control on braking
2. This technique is more efficient during stop and go services mainly in urban regions
3. Prevents wear and tear on mechanical braking
4. It has better fuel efficiency

II. REGENERATIVE BRAKING SYSTEM

A. Working Principle

While driving a vehicle, the wheels produces huge amount of kinetic energy. This energy gets wasted as heat and friction during the mechanical braking. The regenerative braking technique is an energy extraction process in which this kinetic energy is converted into useful electrical energy and can be stored in the batteries. The Fig.1 shows the normal forward driving condition in which the motor moves in forward direction thus torque acts in wheel spinning direction.

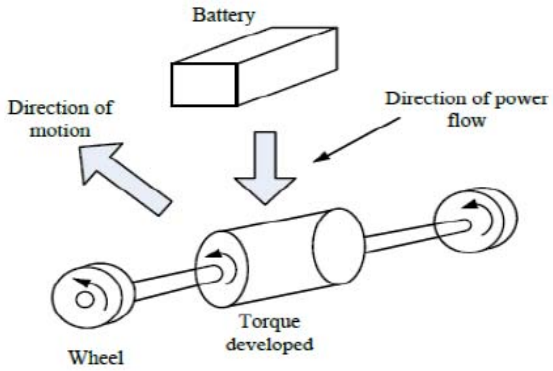


Fig.1. Forward driving condition[4]

The back emf produced opposes the motion of the vehicles according to Lenz law. In this the strength of magnetic field is increasing thus increasing the vehicles speed profile. While, Fig.2 shows the regeneration during braking in which the motor acts as generator thus the strength of magnetic field is reducing and thus reducing the speed. In the motor torque acts in opposite direction to wheel spinning direction. Also, it shows the energy is transferred back to the battery, whereas, above diagram shows that the battery is supplying energy.

Regenerative braking vastly reduces the dependency on fuel, increases fuel conservation and reduces emission. This system works more efficiently in urban areas with stop and go driving conditions. Also, the braking time will be less as compared to conventional braking *i.e*. the vehicle will come to halt in less time.

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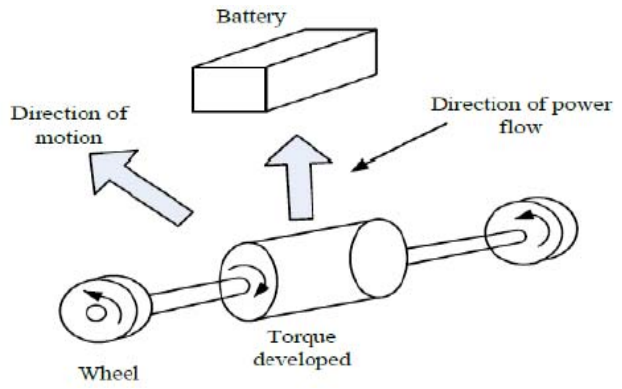


Fig.2. Regenerative braking condition[4]

B. Converter Circuit

The power electronics advances have increased the converter conversion efficiency. The converter circuit works in bidirectional mode during both motoring and regeneration. The switching sequence may vary for motoring and regeneration depending on the input from the controller and brake switch signal. The switching devices may vary depending on the application. Mostly MOSFET switches are used due to its higher switching frequency and suitable application.

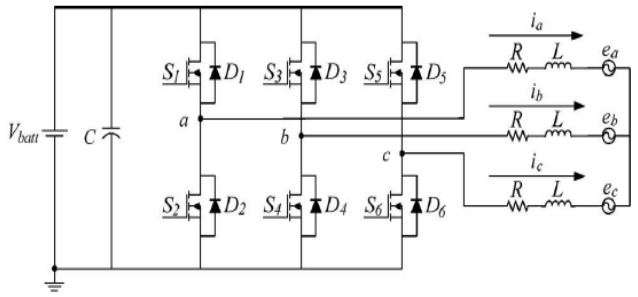


Fig. 3. Converter circuit with BLDC motor load[1]

The above Fig.3. shows the converter circuit realized using MOSFET switches. The RLE load represents the motor stator winding resistance, inductance and the back emf. The stator phase currents are represented by i_a, i_b and i_c. The e_a, e_b and e_c are the per phase back emf for a, b and c phases. The switching sequence is generated by considering the hall sensor output for each 120° phase shift.

The gate pulses are generated by the controller by considering the load current requirement and produces duty cycle accordingly. During motoring, the high side switches are given pulse width modulation(PWM) pulses, while low side are given normal switching pulses. This helps in reducing the switching loss at lower half of switches. For energy regeneration mode both higher and low side switches are given PWM pulses.

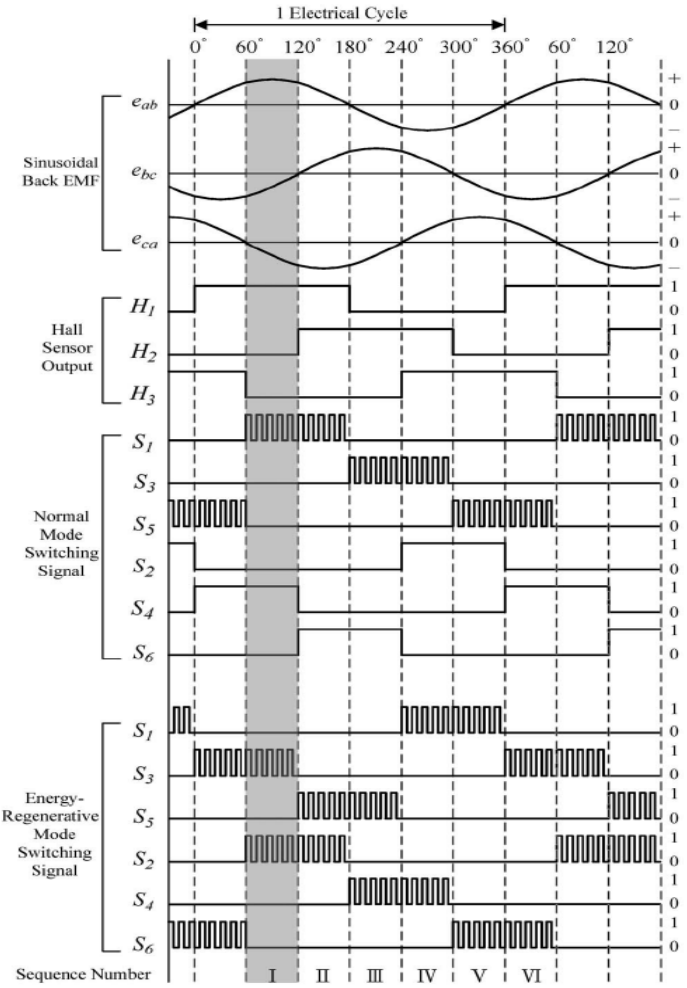
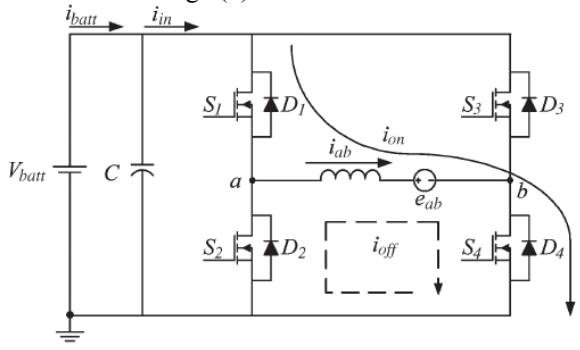


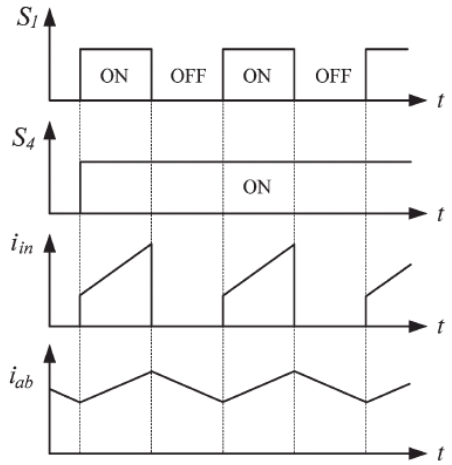
Fig.4. waveforms for hall sensor output, back emf, and switching signal for normal mode and regenerative mode[1]

1. Normal Mode

During the switching on of S_1 and S_4, the energized loop current i_{on} increases the current i_{ab} through the winding inductor. As i_{ab} increases back emf gets generated to oppose the change in magnetic field. In freewheeling mode switch S_1 is switched off, thus the current through inductor will follow the path S_4 and body diode D_2. This makes the discharging path for for i_{off}. The current waveforms during on and off phase are shown in Fig.5(b)



(a)

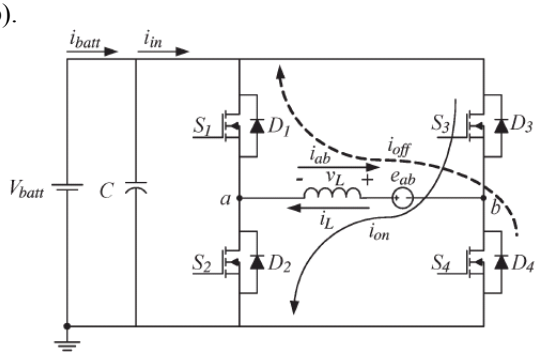


(b)

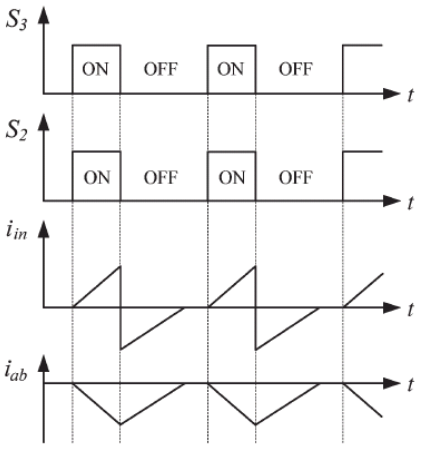
Fig.5. Normal mode. (a) Equivalent Circuit. (b) phase current and input current waveforms [3]

2. Energy Regeneration Mode

The back emf e_{ab} acts as source of voltage during regeneration mode. When S_2 and $S_3$are switched on, the voltage appearing across the inductor will be $V_{batt} + e_{ab}$, because of energizing of winding and the incoming current equals $-i_{ab}$ .During switching off of switches S_2 and S3, freewheeling diodes D_1 and D_4 conducts and provides path for current i_{off} back to battery. The corresponding waveforms are as shown in Fig.6(b).



(a)



(b)

Fig.6. Regeneration mode (a) Equivalent circuit. (b) phase current and input current waveforms[3]

C. Advance Boost Converter Circuit

The condition for regeneration to happen is that, the back emf should be more than the supply voltage. In this case only the energy is fed back into the battery or reverse current will flow from motor back to the supply. At lower speeds the motor is not capable of producing this much back emf. Hence, we have to boost the back emf more than supply voltage. This can be done with a separate boost converter or by using the same converter for boosting by changing the switching of IGBT. Fig.8 shows the equivalent boost converter circuit. It can be realized by using the lower half switch an upper half body diode.

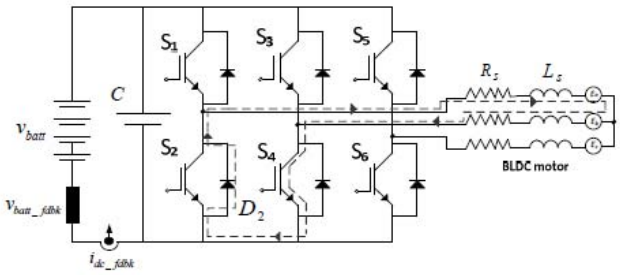


Fig.7. Converter Circuit using IGBT [2]

To provide good electric braking performance, the low side switches S_2 ,S_4 and S_6 are switched accordingly. During turn on of the switch, the inductance of the winding gets charged to a voltage $e_{ab.}$ When switch is turned off, the electric energizes transferred to the battery through top diode.

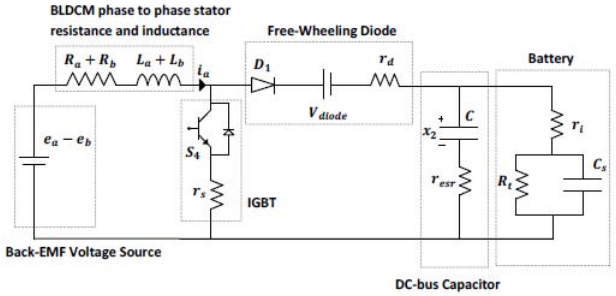


Fig.8. Converter Circuit during boosting operation [2]

III. ENERGY STORAGE SYSTEM

In electric vehicles, the batteries carry the power in the form of electrical energy. The amount of energy stored by it decides the range of vehicle. The batteries state of charge(SOC) is the amount of charge stored by the storage system. The batteries in electric vehicles are defined by their energy density, moderately high power to weight ratio and energy to weight ratio.

The graph in Fig.9 shows the comparative study of various batteries on specific energy density and volumetric energy density parameters. The Lithium-ion battery is has lighter weight and smaller size with highest specific energy density.

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Mostly lead acid battery are used because of their price and reliability. The battery should not be charged above its SOC, If so, it will degrade its performance. But lead acid batteries can sustain voltage above rated without performance degradation.

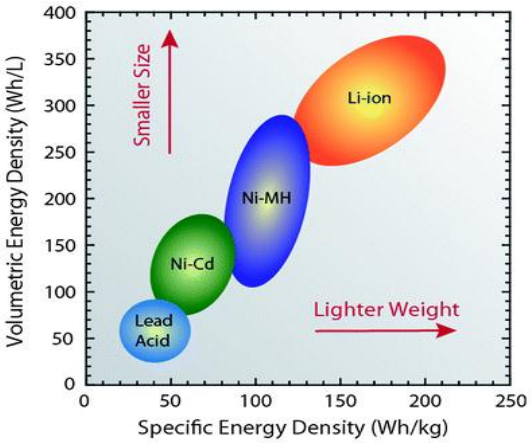


Fig.9. Comparison of different batteries

1. Energy Recovery Mechanism

Energy recovery process is the extraction of kinetic energy from the wheels of vehicle. This energy gets wasted during conventional braking. The regenerative braking is using this wasted energy to convert into useful electrical energy and store in batteries. This in turn helps in extending the range of the vehicle.

The kinetic energy contained in the vehicle is given by,

$$K.E. = \frac{1}{2}Mv^2$$

Where, m is mass of the vehicle and v is the speed with which vehicle is moving.

The amount of energy extracted is maximum when the speed is above rated, during coasting condition or on down slope region. At this condition, the battery voltage may reach above rated value any may harm the battery, So, to limit battery SOC some provision ha to be made to dissipate excess energy when battery gets fully charged. For this, one resistance is connected across dc link, in which excess energy gets wasted as heat. As the speed of engine increases, the losses also increases. The engine losses can be reduced by maintaining lower speed, thus increasing tractive torque and allowing the vehicle to keep on moving for longer duration while coasting. The fuel efficiency can be increased, by increasing the coasting time more often when travelling downhill.

IV. IMPROVEMENT IN SYSTEM

A. Control Strategy

The function of brake controller is to continuously check the wheel speed and estimate the rotational force, torque and produce electricity to supply back to the battery. The improvement in motor control algorithm provides entire control on motor torque in case of both motoring and regeneration.

1. Fuzzy Logic Control

The artificial intelligence has advanced the existing system to function more efficiently. The fuzzy logic control strategy can be easily demonstrated by the influence of different factors for electric vehicles braking force distribution. Therefore, the fuzzy control theory is applied to the EV braking force distribution [10].

Input variables are defined with a membership function(MF). It is a curve that defines how each point in the input space is mapped to a membership value between 0 and 1. The vehicle speed, SOC and required braking force are the preferred three inputs of the fuzzy controller. The output is the regenerative braking force feeding into the machine.

B. Flywheel Implementation

A flywheel is known as Electromechanical battery. It is an inertia energy storage device. Flywheel stores energy when supplied energy is more than required and release it when energy demand is more than supply.

The reason why flywheels are used are:

- Flexibility in design and operation
- Higher cycle efficiency up to 90%
- Although in fast charging profile it has low effect on life cycle
- More than 10 years of long maintenance period.

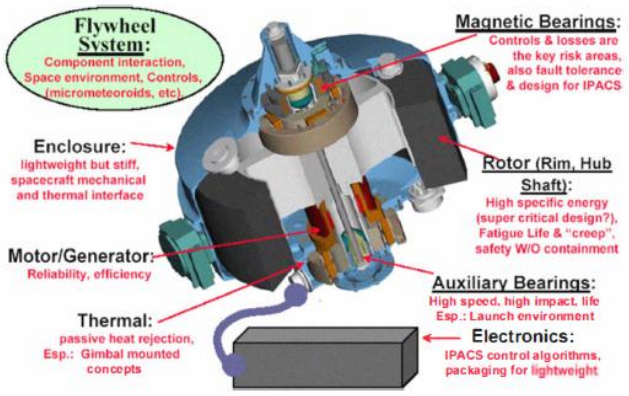


Fig.10. Flywheel System[1]

C. *Use of Ultracapacitor*

The ultracapacitor is a high-capacity capacitor having capacitance values much higher than other capacitors. It is having properties intermediate between electrolytic capacitor rechargeable batteries. The use of ultracapacitor in electric vehicle provides additional storage space which in turn helps in extending the range of the vehicle.

Advantages of using Ultracapacitor such as[1]:

- Improves EV's transient performance
- Compared to electrolytic capacitors it stores 20% more energy
- During acceleration and braking it avoids fast and sudden battery discharge

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In this system, Aluminium Smoothing inductor Ls, DC-DC converter using IGBT switch, Ultracapacitor bank , and battery pack are the four main components. During the charging phase boosting action will happen by turning on the

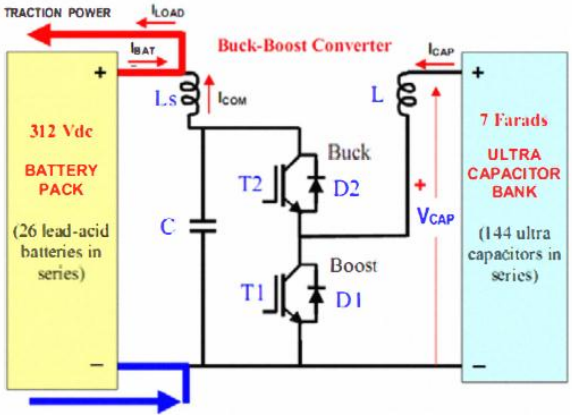


Fig.11. Buck- Boost Converter with Ultracapacitor Bank across dc link [1]

Switch T_1, so transferring energy from ultracapacitor bank to battery. During discharging of battery, switch T_2 is switched on supplying energy to the ultracapacitor bank. Ultracapacitor acts as intermediate stage between the battery and the converter, thus adding on excess energy storage.

V. CONCLUSION

This paper summarizes how the regenerative braking technique employed for electric vehicle helps in improving the efficiency and driving range. The use of advance power electronics helps in improving the efficiency of conversion. The improvement in system can be done by using flywheel and ultracapacitor. The regenerative braking improves the driving range of 16.25%.

This paper also discusses the need of boosting the voltage when back emf is not sufficient to provide reverse current to battery, especially at lower speeds. The advance control algorithms like fuzzy logic improves the energy savings in electric vehicle.

FUTURE SCOPE

Further research is needed in regenerative braking to create a better system for extracting more energy and reduce the braking time. The use of artificial intelligence can bring smartness to the control. Researchers are working on regeneration in fuel cells *i.e* unitized regenerative fuel cell(URFC) to have system with light weight, good energy density and cost effective.

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